Engineering Mathematics

Detailed Solutions

GATE ENGINEERING MATHEMATICS

(EC, EE, IN, CS, ME, CE, CH, PI)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!Series

Preface

Welcome to PrepFusion™!

This book works, in full, every question GATE has actually asked in Transform Theory (EC/EE/IN/CS/ME/CE/CH/PI), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, and every solution is taken step by step rather than asserted.

Each solution carries the same identifier as the question it answers in the companion questions volume, so you can move between practice and explanation without a lookup table. Where the examining authority revised an answer or awarded marks to all (MTA), that is noted with the question it affects.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

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Transform Theory

Engineering Mathematics

SOLUTIONS — Transform Theory

ME

Mechanical Engineering

Q8.26.1 MCQ 2026 2M Ans: A ● ● ○

We need to find the Laplace transform of the trigonometric product function:

$$f(t) = \sin(2t)\sin(4t)$$

Let us use the product-to-sum trigonometric identity to rewrite $f(t)$:

$$\sin(A)\sin(B) = \frac{1}{2}\Big[\cos(A-B) - \cos(A+B)\Big]$$

Substituting $A = 4t$ and $B = 2t$:

$$f(t) = \frac{1}{2}\Big[\cos(4t-2t) - \cos(4t+2t)\Big] = \frac{1}{2}\cos(2t) - \frac{1}{2}\cos(6t)$$

Now, applying the standard Laplace transform formula for cosine terms, $\mathcal{L}\{\cos(\omega t)\} = \frac{s}{s^2+\omega^2}$:

$$\begin{aligned}\mathcal{L}\{f(t)\} &= \frac{1}{2}\mathcal{L}\{\cos(2t)\} - \frac{1}{2}\mathcal{L}\{\cos(6t)\} \\ &= \frac{1}{2}\left(\frac{s}{s^2+2^2}\right) - \frac{1}{2}\left(\frac{s}{s^2+6^2}\right) \\ &= \frac{s}{2}\left[\frac{1}{s^2+4} - \frac{1}{s^2+36}\right]\end{aligned}$$

Combine the terms inside the brackets over a common denominator:

$$\begin{aligned}\mathcal{L}\{f(t)\} &= \frac{s}{2}\left[\frac{(s^2+36)-(s^2+4)}{(s^2+4)(s^2+36)}\right] \\ &= \frac{s}{2}\left[\frac{32}{(s^2+4)(s^2+36)}\right] \\ &= \frac{16s}{(s^2+4)(s^2+36)}\end{aligned}$$

This corresponds to choice A.

$$\boxed{\text{A. } \frac{16s}{(s^2+4)(s^2+36)}}$$

Key Points

- Simplify products of sine or cosine functions using product-to-sum identities before applying linearity and standard Laplace transform pairs.

Q8.23.1 MCQ 2023 2M Ans: A ● ● ○

We need to find the inverse Laplace transform of the rational function:

$$F(s) = \frac{1}{s^3 - s}$$

First, factor the denominator completely:

$$s^3 - s = s(s^2-1) = s(s-1)(s+1)$$

Now, expand $F(s)$ into partial fractions:

$$\frac{1}{s(s-1)(s+1)} = \frac{A}{s} + \frac{B}{s-1} + \frac{C}{s+1}$$

Using the cover-up method to find the coefficients:

- For A (at $s = 0$): $A = \dfrac{1}{(0-1)(0+1)} = -1$
- For B (at $s = 1$): $B = \dfrac{1}{1(1+1)} = \dfrac{1}{2}$
- For C (at $s = -1$): $C = \dfrac{1}{-1(-1-1)} = \dfrac{1}{2}$

Substitute these coefficients back into the partial fraction expansion:

$$F(s) = -\frac{1}{s} + \frac{1}{2}\left(\frac{1}{s-1}\right) + \frac{1}{2}\left(\frac{1}{s+1}\right)$$

Apply the inverse Laplace transform term-by-term using standard pairs ($\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = 1$, $\mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}$):

$$f(t) = \left(-1 + \frac{1}{2}e^{t} + \frac{1}{2}e^{-t}\right)u(t)$$

This corresponds to choice A.

$$\boxed{\text{A. } \left(-1 + \frac{1}{2}e^{-t} + \frac{1}{2}e^{t}\right)u(t)}$$

Key Points

- Resolve rational functions with distinct real poles into linear partial fractions to quickly evaluate term-by-term exponential inverse transforms.

Q8.22.1 MCQ 2022 1M Ans: B ● ● ○

We are given a periodic square wave function $F(t)$ that alternates between values of 4 and 0 every 1 second. We need to find the constant term (a_0) in its Fourier series expansion.

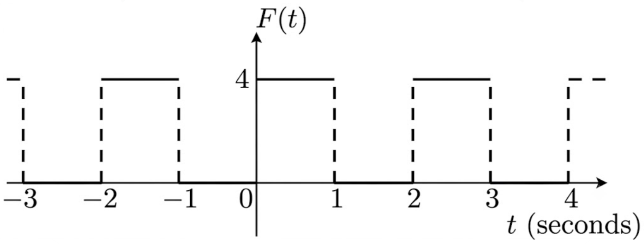

Fig. Solution Q8.22.1

Let us determine the total period T of the wave from the provided graph:

- The function stays at 4 for 1 second and at 0 for 1 second. Thus, the fundamental time period is $T = 1 + 1 = 2$ seconds.

The constant term a_0 in a standard Fourier series represents the average value of the function over one full period:

$$a_0 = \frac{1}{T}\int_0^T F(t)\,dt$$

Since $F(t) = 4$ for $t \in [0, 1)$ and $F(t) = 0$ for $t \in [1, 2)$:

$$a_0 = \frac{1}{2}\left[\int_0^1 4\,dt + \int_1^2 0\,dt\right] = \frac{1}{2}\Big[4(1-0) + 0\Big] = \frac{4}{2} = 2$$

This corresponds to choice B.

$$\boxed{\text{B. } 2}$$

Key Points

- The constant term or DC offset in any Fourier series expansion is simply the geometric mean value of the periodic signal evaluated over one full cycle.

Q8.22.2 MCQ 2022 1M Ans: A ● ● ○

We need to find the nature of the Fourier series expansion of $f(x) = x^3$ defined in the symmetric interval $[-1, 1]$. Let us evaluate the parity of the function $f(x) = x^3$:

$$f(-x) = (-x)^3 = -x^3 = -f(x)$$

Since $f(-x) = -f(x)$, $f(x)$ is a strictly **odd function**. Let us analyze the standard Fourier expansion properties for odd functions over a symmetric interval $[-L, L]$:

- The constant offset term $a_0 = 0$.
- The cosine coefficients $a_n = 0$ because the product of an odd function and an even function ($\cos(n\pi x/L)$) is odd, which integrates to zero over a symmetric interval.
- The sine coefficients $b_n \neq 0$ because the product of two odd functions ($\sin(n\pi x/L)$) creates an even function, which yields a non-zero integral.

Therefore, the series contains only sine terms. This matches choice A.

$$\boxed{\text{A. only sine terms}}$$

Key Points

- **Fourier Parity Rule**: Odd functions expanded over symmetric intervals have zero cosine coefficients ($a_0 = 0, a_n = 0$), meaning their Fourier series expansions consist entirely of sine terms.

Q8.15.1 MCQ 2015 1M Ans: A ● ○ ○

We need to state the standard canonical Laplace transform of the function:

$$f(t) = \cos(\omega t)$$

By standard definition of the unilateral Laplace integral:

$$\mathcal{L}\{\cos(\omega t)\} = \int_0^\infty e^{-st} \cos(\omega t)\, dt = \frac{s}{s^2 + \omega^2}$$

This corresponds to choice A.

$$\boxed{\text{A. } \frac{s}{s^2 + \omega^2}}$$

Key Points

- The Laplace transform of $\cos(\omega t)$ maps directly to $\dfrac{s}{s^2 + \omega^2}$ for $\text{Re}(s) > 0$.

Q8.15.2 MCQ 2015 1M Ans: B ● ● ○

We need to find the Laplace transform of the complex exponential function $f(t) = e^{5it}$, where $i = \sqrt{-1}$.
Using the standard exponential transform pair $\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$ where $a = 5i$:

$$\mathcal{L}\{e^{5it}\} = \frac{1}{s-5i}$$

Multiply the numerator and denominator by the complex conjugate of the denominator $(s + 5i)$:

$$\mathcal{L}\{e^{5it}\} = \frac{1}{s-5i} \times \frac{s+5i}{s+5i} = \frac{s+5i}{s^2-(5i)^2} = \frac{s+5i}{s^2+25}$$

This corresponds to choice B.

$$\boxed{\text{B. } \frac{s+5i}{s^2+25}}$$

Key Points

- Complex exponential transforms can be resolved into standard rational fractions by multiplying both the numerator and denominator by the complex conjugate of the pole.

Q8.15.3 MCQ 2015 1M Ans: C ● ● ○

We are given a rectangular gate pulse function $f(t)$ shown ;

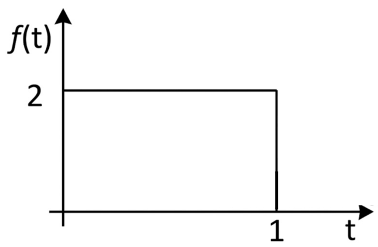

Fig. Solution Q8.15.3

$$f(t) = \begin{cases} 2 & \text{if } 0 \le t < 1 \\ 0 & \text{if } t \ge 1 \end{cases}$$

We need to find its Laplace transform.
Let us express this step pulse algebraically using unit step functions $(u(t))$:

$$f(t) = 2\left[u(t) - u(t-1)\right]$$

Now, apply the time-shifting property of the Laplace transform ($\mathcal{L}\{u(t-c)\} = \frac{e^{-cs}}{s}$):

$$\mathcal{L}\{f(t)\} = 2\left(\frac{1}{s}\right) - 2\left(\frac{e^{-1\cdot s}}{s}\right) = \frac{2-2e^{-s}}{s}$$

This matches choice C perfectly.

$$\boxed{\text{C. } \frac{2-2e^{-s}}{s}}$$

Key Points

- A rectangular pulse spanning from $t = 0$ to $t = c$ with a height of A is expressed as $A[u(t) - u(t-c)]$. Its Laplace transform evaluates directly to $\frac{A(1-e^{-cs})}{s}$.

Q8.15.4 MCQ 2015 1M Ans: A ● ● ○

We need to find the Laplace transform of the modulated sine function:

$$f(t) = e^{2t}\sin(5t)u(t)$$

Let us use the First Shifting Theorem, which states that if $\mathcal{L}\{g(t)\} = G(s)$, then $\mathcal{L}\{e^{at}g(t)\} = G(s-a)$. First, find the Laplace transform of the core sine term $g(t) = \sin(5t)$:

$$G(s) = \mathcal{L}\{\sin(5t)\} = \frac{5}{s^2+5^2} = \frac{5}{s^2+25}$$

Now, apply the exponential shift multiplier e^{2t}, replacing every s with $(s-2)$:

$$\mathcal{L}\{e^{2t}\sin(5t)\} = G(s-2) = \frac{5}{(s-2)^2+25} = \frac{5}{s^2-4s+29}$$

This corresponds to choice A.

$$\boxed{\text{A. } \frac{5}{s^2-4s+29}}$$

Key Points

- Multiplying a time function by an exponential term e^{at} shifts its Laplace transform in the frequency domain, substituting $s \to (s-a)$.

Q8.14.1 MCQ 2014 1M Ans: D ● ● ○

We are given that $\mathcal{L}\{\cos(\omega t)\} = \frac{s}{s^2+\omega^2}$. We need to find the Laplace transform of:

$$f(t) = e^{-2t}\cos(4t)$$

Let us apply the First Shifting Theorem. Since the exponential factor is e^{-2t} (so $a = -2$), we substitute $s \to (s-(-2)) = (s+2)$. First, find the Laplace transform of the core cosine term with $\omega = 4$:

$$G(s) = \mathcal{L}\{\cos(4t)\} = \frac{s}{s^2+4^2} = \frac{s}{s^2+16}$$

Now, apply the shift substitution $s \to (s+2)$:

$$\mathcal{L}\{e^{-2t}\cos(4t)\} = \frac{s+2}{(s+2)^2+16}$$

This matches choice D perfectly.

$$\boxed{\text{D. } \frac{s+2}{(s+2)^2+16}}$$

Key Points

- Frequency domain translation shifts the variable s based on the exponential coefficient: $\mathcal{L}\{e^{-at}\cos(\omega t)\} = \dfrac{s+a}{(s+a)^2+\omega^2}$.

Q8.13.1 MCQ 2013 1M Ans: C ● ● ○

We are given a second-order linear differential equation:

$$\frac{d^2 f}{dt^2} + f = 0$$

with initial conditions $f(0) = 0$ and $\frac{df}{dt}(0) = 4$. We need to find the Laplace transform $F(s) = \mathcal{L}\{f(t)\}$.
Let us take the Laplace transform of both sides of the differential equation:

$$\mathcal{L}\left\{\frac{d^2 f}{dt^2}\right\} + \mathcal{L}\{f(t)\} = 0$$

Using the derivative property of the Laplace transform:

$$\left[s^2 F(s) - s \cdot f(0) - \frac{df}{dt}(0)\right] + F(s) = 0$$

Substitute the initial conditions ($f(0) = 0$ and $f'(0) = 4$):

$$s^2 F(s) - 4 + F(s) = 0 \implies F(s)(s^2+1) = 4$$

$$F(s) = \frac{4}{s^2+1}$$

This corresponds to choice C.

$$\boxed{\text{C. } \frac{4}{s^2+1}}$$

Key Points

- Differential equations can be converted into simple algebraic expressions by applying the derivative property of the Laplace transform and substituting the initial boundary condition values.

Q8.12.1 MCQ 2012 1M Ans: D ● ● ○

We need to find the inverse Laplace transform of the function:

$$F(s) = \frac{1}{s(s+1)}$$

Let us expand $F(s)$ into partial fractions:

$$F(s) = \frac{1}{s} - \frac{1}{s+1}$$

Now, apply the inverse Laplace transform term-by-term using standard pairs ($\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = 1$, $\mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\} = e^{-t}$):

$$f(t) = 1 - e^{-t}$$

This corresponds to choice D.

$$\boxed{\text{D. } f(t) = 1 - e^{-t}}$$

Key Points

- Rational functions with simple roots in the denominator can be split into linear partial fractions to quickly evaluate the inverse transform term-by-term.

Q8.10.1 MCQ 2010 1M Ans: A ● ● ○

We are given the Laplace transform of a function:

$$F(s) = \frac{1}{s^2(s+1)}$$

We need to find the time domain function $f(t)$. Let us use the partial fraction expansion method:

$$F(s) = -\frac{1}{s} + \frac{1}{s^2} + \frac{1}{s+1}$$

Apply the inverse Laplace transform term-by-term using standard pairs ($\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = 1$, $\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} = t$, $\mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\} = e^{-t}$):

$$f(t) = t - 1 + e^{-t}$$

This matches choice A perfectly.

$$\boxed{\text{A. } t - 1 + e^{-t}}$$

Key Points

- A repeated pole at the origin (s^2) can be solved cleanly using a partial fraction expansion or by integrating the core inverse transform $1 - e^{-t}$ with respect to time.

Q8.07.1 MCQ 2007 1M Ans: A ● ● ○

We are given that $\mathcal{L}\{f(t)\} = F(s)$. We need to find the Laplace transform of the integral function:

$$g(t) = \int_0^t f(x)\,dx$$

Let us utilize the standard integration property of the Laplace transform:

$$\mathcal{L}\left\{\int_0^t f(x)\,dx\right\} = \frac{1}{s}F(s)$$

This matches choice A perfectly.

$$\boxed{\text{A. } \frac{1}{s}F(s)}$$

Key Points

- **Integration Property**: Integrating a time-domain function from 0 to t is equivalent to dividing its Laplace transform by s in the frequency domain.

Q8.99.1 MCQ 1999 1M Ans: C ● ● ○

We need to find the Laplace transform of the binomial function:

$$f(t) = (a + bt)^2 = a^2 + 2abt + b^2t^2$$

where a and b are constants.
Apply the linearity property of the Laplace transform to evaluate each term independently:

$$\mathcal{L}\{f(t)\} = a^2\mathcal{L}\{1\} + 2ab\mathcal{L}\{t\} + b^2\mathcal{L}\{t^2\}$$

Using the standard polynomial transform pair $\mathcal{L}\{t^n\} = \dfrac{n!}{s^{n+1}}$:

$$\mathcal{L}\{(a + bt)^2\} = a^2\left(\frac{1}{s}\right) + 2ab\left(\frac{1}{s^2}\right) + b^2\left(\frac{2}{s^3}\right) = \frac{a^2}{s} + \frac{2ab}{s^2} + \frac{2b^2}{s^3}$$

This corresponds to choice C.

$$\boxed{\text{C. } (a^2/s) + (2ab/s^2) + (2b^2/s^3)}$$

Key Points

- Expand non-linear polynomial expressions into distinct additive power terms before applying linearity and the standard power transform pair $\mathcal{L}\{t^n\} = \dfrac{n!}{s^{n+1}}$.

Q8.94.1 MCQ 1994 1M Ans: A ● ● ○

We need to find the Laplace transform of the hyperbolic cosine function:

$$f(t) = \cosh(mt)$$

Let us use the exponential definition of the hyperbolic cosine function:

$$\cosh(mt) = \frac{e^{mt} + e^{-mt}}{2}$$

Now, apply the linearity property of the Laplace transform:

$$\mathcal{L}\{\cosh(mt)\} = \frac{1}{2}\mathcal{L}\{e^{mt}\} + \frac{1}{2}\mathcal{L}\{e^{-mt}\} = \frac{1}{2}\left(\frac{1}{s-m}\right) + \frac{1}{2}\left(\frac{1}{s+m}\right)$$

Combine these fractions over a common denominator:

$$\mathcal{L}\{\cosh(mt)\} = \frac{1}{2}\left[\frac{(s+m) + (s-m)}{s^2 - m^2}\right] = \frac{s}{s^2 - m^2}$$

$$\boxed{A : \frac{s}{s^2 - m^2}}$$

Key Points

- The Laplace transform of the hyperbolic cosine function $\cosh(mt)$ evaluates directly to $\dfrac{s}{s^2 - m^2}$ for $\mathrm{Re}(s) > |m|$.

Q8.93.1 MCQ | 1993 | 1M | Ans: B | ● ● ○

We are given a periodic function $f(t)$ shown . The function completes one full cycle over the interval $[0, 2\pi]$, so the period is $T = 2\pi$. Within this first period:

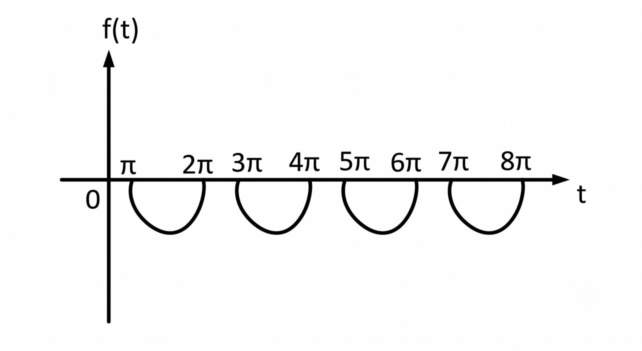

Fig. Solution Q8.93.1

- $f(t) = 0$ for $t \in [0, \pi)$
- $f(t) = \sin(t)$ for $t \in [\pi, 2\pi)$

For any periodic function with a fundamental period T, the Laplace transform is given by:

$$\mathcal{L}\{f(t)\} = \frac{1}{1 - e^{-Ts}} \int_0^T f(t)e^{-st}\, dt$$

Evaluating the single-period integral component:

$$I_1 = \int_\pi^{2\pi} \sin(t)e^{-st}\, dt = \left[\frac{e^{-st}}{s^2+1}\Big(-s\sin(t) - \cos(t)\Big)\right]_\pi^{2\pi} = -\frac{e^{-\pi s}(1 + e^{-\pi s})}{s^2+1}$$

Now, substitute I_1 back into the periodic transform formula and use the difference of squares factorization $1 - e^{-2\pi s} = (1 - e^{-\pi s})(1 + e^{-\pi s})$ to simplify the expression:

$$\mathcal{L}\{f(t)\} = \frac{1}{(1 - e^{-\pi s})(1 + e^{-\pi s})} \cdot \left[-\frac{e^{-\pi s}(1 + e^{-\pi s})}{s^2+1}\right] = -\frac{1}{1+s^2} \cdot \frac{e^{-\pi s}}{1 - e^{-\pi s}}$$

$$\boxed{\text{B: } \frac{e^{-\pi s}}{(1+s^2)(e^{-\pi s} - 1)}}$$

SOLUTIONS — Transform Theory

CE

Civil Engineering

Q8.25.1 MCQ | 2025 | 1M | Ans: C | ● ● ○

We are given a periodic function $f(x)$ with period $T = 4$, defined over one cycle as:

$$f(x) = \begin{cases} 2k & \text{if } -1 < x < 1 \\ 0 & \text{if } -2 < x < -1 \text{ or } 1 < x < 2 \end{cases}$$

We need to determine its correct Fourier series expansion.

Step 1: Analyze function parity (Symmetry) Let us check if the function is even or odd by evaluating $f(-x)$ within the non-zero region: For $x \in (-1, 1)$, $-x$ also lies in $(-1, 1)$, so $f(-x) = 2k = f(x)$. Since $f(-x) = f(x)$, $f(x)$ is a strictly even function.

Because it is an even function, all sine coefficients vanish completely ($b_n = 0$). The Fourier series will consist exclusively of a constant term and cosine terms, eliminating choices B and D.

Step 2: Calculate the constant term (a_0) The period is $T = 4$, so the half-period is $L = \frac{T}{2} = 2$. For an even function:

$$a_0 = \frac{1}{L}\int_{-L}^{L} f(x)\,dx = \frac{2}{L}\int_0^L f(x)\,dx$$

$$a_0 = \frac{2}{2}\int_0^1 2k\,dx = \Big[2kx\Big]_0^1 = 2k$$

The constant base term in the standard Fourier series formula is $\frac{a_0}{2}$:

$$\text{Constant Term} = \frac{a_0}{2} = \frac{2k}{2} = k$$

This matches choices C and D, ruling out choice A.

Step 3: Calculate the cosine coefficients (a_n)

$$a_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)dx = \frac{2}{2}\int_0^1 2k\cos\left(\frac{n\pi x}{2}\right)dx$$

$$a_n = 2k\left[\frac{\sin\left(\frac{n\pi x}{2}\right)}{\frac{n\pi}{2}}\right]_0^1 = \frac{4k}{n\pi}\sin\left(\frac{n\pi}{2}\right)$$

Let us evaluate a_n for sequential integer values of n:

- For even n ($n = 2, 4, 6, \dots$): $\sin(n\pi) = 0 \implies a_n = 0$
- For $n = 1$: $a_1 = \frac{4k}{\pi}\sin\left(\frac{\pi}{2}\right) = \frac{4k}{\pi}$
- For $n = 3$: $a_3 = \frac{4k}{3\pi}\sin\left(\frac{3\pi}{2}\right) = -\frac{4k}{3\pi}$
- For $n = 5$: $a_5 = \frac{4k}{5\pi}\sin\left(\frac{5\pi}{2}\right) = \frac{4k}{5\pi}$

Substituting these non-zero coefficients back into the expansion yields:

$$f(x) = k + \frac{4k}{\pi}\cos\left(\frac{\pi}{2}x\right) - \frac{4k}{3\pi}\cos\left(\frac{3\pi}{2}x\right) + \frac{4k}{5\pi}\cos\left(\frac{5\pi}{2}x\right) - \dots$$

$$f(x) = k + \frac{4k}{\pi}\left(\cos\frac{\pi}{2}x - \frac{1}{3}\cos\frac{3\pi}{2}x + \frac{1}{5}\cos\frac{5\pi}{2}x - \dots\right)$$

This matches choice C perfectly.

$$\boxed{\text{C: } f(x) = k + \frac{4k}{\pi}\left(\cos\frac{\pi}{2}x - \frac{1}{3}\cos\frac{3\pi}{2}x + \frac{1}{5}\cos\frac{5\pi}{2}x - \dots\right)}$$

Key Points

- Symmetric step functions centered at the origin are even, eliminating all sine terms ($b_n = 0$). Calculating the baseline average value quickly helps narrow down the correct multiple-choice options.

Q8.22.1 NAT 2022 1M Ans: 0.120-0.130

The Fourier cosine series of a function is defined as:

$$f(x) = \sum_{n=0}^{\infty} f_n \cos(nx)$$

We are given the function $f(x) = \cos^4 x$. We need to find the numerical value of the combined coefficients sum $(f_4 + f_5)$.
Let us expand $f(x) = \cos^4 x$ into individual harmonic frequencies using standard trigonometric multiple-angle power reduction identities:

$$\cos^2 x = \frac{1 + \cos(2x)}{2}$$

$$\cos^4 x = \left(\cos^2 x\right)^2 = \left(\frac{1 + \cos(2x)}{2}\right)^2$$
$$= \frac{1}{4}\left[1 + 2\cos(2x) + \cos^2(2x)\right]$$

Now, apply the power reduction identity to the $\cos^2(2x)$ term:

$$\cos^2(2x) = \frac{1 + \cos(4x)}{2}$$

$$\cos^4 x = \frac{1}{4}\left[1 + 2\cos(2x) + \frac{1 + \cos(4x)}{2}\right]$$
$$= \frac{1}{4}\left[1 + 2\cos(2x) + \frac{1}{2} + \frac{1}{2}\cos(4x)\right]$$
$$= \frac{1}{4}\left[\frac{3}{2} + 2\cos(2x) + \frac{1}{2}\cos(4x)\right]$$
$$= \frac{3}{8} + \frac{1}{2}\cos(2x) + \frac{1}{8}\cos(4x)$$

Now, let us match these terms directly with the generalized coefficients f_n from the Fourier cosine series definition:

- Constant term: $f_0 = \frac{3}{8}$
- Second harmonic: $f_2 = \frac{1}{2}$
- Fourth harmonic: $f_4 = \frac{1}{8} = 0.125$
- All other harmonic coefficients are zero ($f_1 = f_3 = f_5 = \cdots = 0$).

Therefore, the target coefficients sum evaluates to:

$$f_4 + f_5 = 0.125 + 0 = 0.125$$

$$\boxed{0.125}$$

Key Points

- Pure trigonometric polynomials are expanded into their equivalent Fourier series representations directly using multiple-angle algebraic identities, bypassing the need for full integral computations.

Q8.11.1 | MCQ | 2011 | 1M | Ans: A | ● ● ○

Given two continuous-time signals $x(t) = e^{-t}$ and $y(t) = e^{-2t}$ defined for $t > 0$, we need to find their convolution:

$$z(t) = x(t) * y(t)$$

By the formal definition of the continuous-time convolution integral for causal signals ($t > 0$):

$$z(t) = \int_0^t x(\tau)y(t-\tau)\,d\tau$$

Substitute the algebraic expressions for $x(t)$ and $y(t)$ into the integral:

$$z(t) = \int_0^t e^{-\tau}e^{-2(t-\tau)}\,d\tau = \int_0^t e^{-\tau}e^{-2t}e^{2\tau}\,d\tau$$

Since the integration is with respect to τ, we can factor out the constant exponential term e^{-2t}:

$$z(t) = e^{-2t}\int_0^t e^{\tau}\,d\tau$$

Evaluate the simple exponential integral:

$$z(t) = e^{-2t}\left[e^{\tau}\right]_0^t = e^{-2t}\left(e^t - e^0\right) = e^{-2t}\left(e^t - 1\right)$$

Distribute the outer e^{-2t} term:

$$z(t) = e^{-2t}\cdot e^t - e^{-2t}\cdot 1 = e^{-t} - e^{-2t}$$

This corresponds to choice A.

$$\boxed{\text{A. } e^{-t} - e^{-2t}}$$

Key Points

- Bounded convolution integrals for causal exponential signals can be evaluated either through direct time-domain integration or by mapping the signals to the s-domain via Laplace transforms ($\mathcal{L}\{x * y\} = X(s)\cdot Y(s)$).

Q8.11.2 | MCQ | 2011 | 2M | Ans: B | ● ● ○

We are given the Laplace transform of a continuous time function $f(t)$:

$$F(s) = \frac{2(s+1)}{s^2+4s+7}$$

We need to find the initial value $f(0)$ and the final value $f(\infty)$ of $f(t)$, respectively. **Step 1: Calculate the initial value using the Initial Value Theorem (IVT)** The Initial Value Theorem states that $f(0) = \lim_{s\to\infty} sF(s)$:

$$f(0) = \lim_{s\to\infty} s\left[\frac{2(s+1)}{s^2+4s+7}\right] = \lim_{s\to\infty}\frac{2s^2+2s}{s^2+4s+7}$$

Divide the numerator and denominator by the highest power s^2:

$$f(0) = \lim_{s\to\infty} \frac{2 + \frac{2}{s}}{1 + \frac{4}{s} + \frac{7}{s^2}} = \frac{2+0}{1+0+0} = 2$$

Step 2: Calculate the final value using the Final Value Theorem (FVT) First, let us check the stability of the system by finding the roots of the denominator polynomial (poles) to ensure the FVT is valid:

$$s^2 + 4s + 7 = 0 \implies s = \frac{-4 \pm \sqrt{16-28}}{2} = -2 \pm i\sqrt{3}$$

Since both poles have strictly negative real parts ($\text{Re}(s) = -2 < 0$), the system is asymptotically stable, and its transient response decays to zero as $t \to \infty$. Therefore, the Final Value Theorem applies:

$$f(\infty) = \lim_{s\to 0} sF(s)$$

$$f(\infty) = \lim_{s\to 0} s\left[\frac{2(s+1)}{s^2+4s+7}\right] = \frac{0\cdot 2(1)}{7} = 0$$

Thus, the initial and final values are **2 and 0**, respectively. This matches choice B.

$$\boxed{\text{B. } 2, 0}$$

Key Points

- **Value Limit Theorems**: The Initial Value Theorem ($\lim_{s\to\infty} sF(s)$) and the Final Value Theorem ($\lim_{s\to 0} sF(s)$) allow us to find the boundary limits of a time signal directly from its frequency-domain transform, provided all system poles lie strictly in the left half of the s-plane.

Q8.09.1 MCQ 2009 1M Ans: B ● ○ ○

We need to state the standard algebraic Laplace transform of the hyperbolic cosine function:

$$f(t) = \cosh(ax)$$

Using the standard exponential definition of the hyperbolic cosine function:

$$\cosh(at) = \frac{e^{at} + e^{-at}}{2}$$

Applying linearity to the unilateral transform pairs ($\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$):

$$\mathcal{L}\{\cosh(at)\} = \frac{1}{2}\left(\frac{1}{s-a}\right) + \frac{1}{2}\left(\frac{1}{s+a}\right) = \frac{1}{2}\left[\frac{(s+a)+(s-a)}{(s-a)(s+a)}\right] = \frac{s}{s^2-a^2}$$

This corresponds to choice B.

$$\boxed{\text{B. } \frac{s}{s^2-a^2}}$$

Key Points

- The Laplace transform maps the hyperbolic function $\cosh(at)$ to $\frac{s}{s^2 - a^2}$ for regions of convergence satisfying $\text{Re}(s) > |a|$.

Q8.05.1 MCQ 2005 2M Ans: A ● ● ○

We need to find the Laplace transform of a phase-shifted cosine function:

$$f(t) = \cos(pt + q)$$

First, let us expand the function using the standard trigonometric sum-of-angles identity:

$$\cos(pt + q) = \cos(pt)\cos(q) - \sin(pt)\sin(q)$$

Since q is a constant phase angle, $\cos(q)$ and $\sin(q)$ act as simple scalar multipliers. Now, apply the linearity property of the Laplace transform:

$$\mathcal{L}\{\vec{f}(t)\} = \cos(q) \cdot \mathcal{L}\{\cos(pt)\} - \sin(q) \cdot \mathcal{L}\{\sin(pt)\}$$

Substitute the standard Laplace transform pairs ($\mathcal{L}\{\cos pt\} = \frac{s}{s^2 + p^2}$ and $\mathcal{L}\{\sin pt\} = \frac{p}{s^2 + p^2}$):

$$\begin{aligned}\mathcal{L}\{\cos(pt + q)\} &= \cos(q)\left(\frac{s}{s^2 + p^2}\right) - \sin(q)\left(\frac{p}{s^2 + p^2}\right)\\ &= \frac{s\cos q - p\sin q}{s^2 + p^2}\end{aligned}$$

This corresponds to choice A.

$$\boxed{\text{A. } \frac{s\cos q - p\sin q}{s^2 + p^2}}$$

Key Points

- Expand phase-shifted trigonometric functions using sum-of-angles identities before applying linearity and standard transform pairs to isolate real and imaginary components cleanly.

Q8.05.2 MCQ 2005 1M Ans: A ● ● ○

We are given the Laplace transform of a function $f(t)$:

$$F(s) = \frac{5s^2 + 23s + 6}{s(s^2 + 2s + 2)}$$

We need to find the steady-state final value of $f(t)$ as $t \to \infty$. Let us apply the Final Value Theorem (FVT), which states that $f(\infty) = \lim_{s\to 0} sF(s)$, provided the system is stable. First, verify stability by checking the denominator poles: The pole at the origin $s = 0$ is accounted for by the FVT multiplier. The remaining roots of $s^2 + 2s + 2 = 0$ are:

$$s = \frac{-2 \pm \sqrt{4 - 8}}{2} = -1 \pm i$$

Since these roots have strictly negative real parts ($\text{Re}(s) = -1 < 0$), the transient responses will decay to zero, making the system stable and validating the use of the theorem.

Now, calculate the final value limit:

$$f(\infty) = \lim_{s\to 0} s\left[\frac{5s^2+23s+6}{s(s^2+2s+2)}\right]$$
$$= \lim_{s\to 0}\frac{5s^2+23s+6}{s^2+2s+2}$$
$$= \frac{5(0)^2+23(0)+6}{0^2+2(0)+2} = \frac{6}{2} = 3$$

This corresponds to choice A.

$\boxed{\text{A. } 3}$

Key Points

- Finding the steady-state limit of a stable transfer function is simplified by canceling the pole at the origin using the s multiplier and evaluating the remaining rational expression at $s = 0$.

Q8.04.1 MCQ 2004 1M Ans: B ● ○ ○

We are given the algebraic definition for a delayed unit step function:

$$u(t-a) = \begin{cases} 0 & \text{if } t < a \\ 1 & \text{if } t \geq a \end{cases}$$

We need to find its Laplace transform representation.
By standard definition of the unilateral Laplace integral:

$$\mathcal{L}\{u(t-a)\} = \int_0^\infty e^{-st}u(t-a)\,dt$$

Since $u(t-a) = 0$ for all $t < a$, the lower limit of the integration shifts from 0 to a:

$$\mathcal{L}\{u(t-a)\} = \int_a^\infty e^{-st}(1)\,dt = \left[\frac{e^{-st}}{-s}\right]_a^\infty = 0 - \left(\frac{e^{-as}}{-s}\right) = \frac{e^{-as}}{s}$$

This corresponds to choice B.

$\boxed{\text{B. } \frac{e^{-as}}{s}}$

Key Points

- **Time-Shifting Property**: Delaying a time-domain signal by a constant a is equivalent to multiplying its standard Laplace transform by an exponential factor e^{-as} in the frequency domain.

Q8.03.1 MCQ 2003 1M Ans: B ● ○ ○

We need to state the standard canonical Laplace transform of the function:

$$f(t) = \sin(at)$$

By standard definition of the unilateral Laplace integral:

$$\mathcal{L}\{\sin(at)\} = \int_0^\infty e^{-st}\sin(at)\,dt$$

Using Euler's exponential definition for the sine function, $\sin(at) = \dfrac{e^{iat} - e^{-iat}}{2i}$:

$$\begin{aligned}\mathcal{L}\{\sin(at)\} &= \frac{1}{2i}\int_0^\infty e^{-st}\left(e^{iat} - e^{-iat}\right) dt \\ &= \frac{1}{2i}\left[\frac{1}{s-ia} - \frac{1}{s+ia}\right] \\ &= \frac{1}{2i}\left[\frac{(s+ia)-(s-ia)}{s^2+a^2}\right] = \frac{1}{2i}\left[\frac{2ia}{s^2+a^2}\right] = \frac{a}{s^2+a^2}\end{aligned}$$

This corresponds to choice B.

$$\boxed{\text{B. } \frac{a}{s^2+a^2}}$$

Key Points

- The Laplace transform maps the sine wave function $\sin(at)$ directly to $\dfrac{a}{s^2+a^2}$ for regions of convergence satisfying $\text{Re}(s) > 0$.

Q8.02.1 MCQ 2002 2M Ans: C ● ● ○

We are given a single half-period sine pulse function $f(t)$ defined as:

$$f(t) = \begin{cases} \sin t & \text{for } 0 \le t \le \pi \\ 0 & \text{for } t > \pi \end{cases}$$

We need to find its Laplace transform for $s > 0$.
Let us rewrite this piecewise function as a continuous expression using unit step functions ($u(t)$):

$$f(t) = \sin(t)u(t) - \sin(t)u(t-\pi)$$

To apply the time-shifting property to the second term, we rewrite $\sin(t)$ to match the delay parameter $(t-\pi)$ using the trigonometric identity $\sin(t) = -\sin(t-\pi)$:

$$f(t) = \sin(t)u(t) + \sin(t-\pi)u(t-\pi)$$

Now, take the Laplace transform of both terms using the shifting theorem ($\mathcal{L}\{g(t-c)u(t-c)\} = e^{-cs}G(s)$):

$$\begin{aligned}\mathcal{L}\{f(t)\} &= \mathcal{L}\{\sin(t)u(t)\} + \mathcal{L}\{\sin(t-\pi)u(t-\pi)\} \\ &= \frac{1}{s^2+1} + e^{-\pi s}\left(\frac{1}{s^2+1}\right) \\ &= \frac{1+e^{-\pi s}}{s^2+1}\end{aligned}$$

This corresponds to choice C.

$$\boxed{\text{C. } \frac{1+e^{-\pi s}}{s^2+1}}$$

Key Points

- Bounded piecewise periodic pulses can be analyzed by breaking them into separate, time-shifted step functions. Adjusting the phase angles allows you to use the shifting theorem to find the transform easily.

Q8.01.1 MCQ 2001 1M Ans: D ● ● ○

We need to find the inverse Laplace transform of the rational function:

$$F(s) = \frac{1}{s^2 + 2s}$$

First, factor the denominator:

$$s^2 + 2s = s(s + 2)$$

Now, expand $F(s)$ into partial fractions:

$$\frac{1}{s(s+2)} = \frac{A}{s} + \frac{B}{s+2}$$

Using the cover-up method to find coefficients A and B:

- For A (at $s = 0$): $A = \dfrac{1}{0+2} = \dfrac{1}{2}$
- For B (at $s = -2$): $B = \dfrac{1}{-2} = -\dfrac{1}{2}$

Substitute these coefficients back into the partial fraction expansion:

$$F(s) = \frac{1}{2}\left(\frac{1}{s}\right) - \frac{1}{2}\left(\frac{1}{s+2}\right)$$

Apply the inverse Laplace transform term-by-term using standard pairs ($\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = 1, \mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\} = e^{-2t}$):

$$f(t) = \frac{1}{2} - \frac{1}{2}e^{-2t} = \frac{1 - e^{-2t}}{2}$$

This corresponds to choice D.

$$\boxed{\text{D. } (1 - e^{-2t})/2}$$

Key Points

- Rational functions with simple real poles can be split into linear partial fractions to quickly evaluate the inverse transform term-by-term using standard exponential pairs.

Q8.00.1 MCQ 2000 2M Ans: D ● ● ○

We need to identify the choice that contains the correct pair of properties for the unilateral Laplace transform.
Let us evaluate the standard properties of the Laplace transform step-by-step:

1. **Derivative Property:** The Laplace transform of the first derivative of a function $f(t)$ is given by:

$$\mathcal{L}\left\{\frac{df}{dt}\right\} = sF(s) - f(0)$$

2. **Integration Property:** The Laplace transform of the integral of a function $f(t)$ from 0 to t is given by:

$$\mathcal{L}\left\{\int_0^t f(\tau)\, d\tau\right\} = \frac{1}{s}F(s)$$

Let us review the mathematical expressions provided in choice D:

$$\mathcal{L}\left[\frac{df}{dt}\right] = sF(s) - f(0); \quad \mathcal{L}\left\{\int_0^t f(\tau)\, d\tau\right\} = \frac{1}{s}F(s)$$

This matches both fundamental transform properties perfectly.

$$\boxed{\text{D}}$$

Key Points

- The derivative property introduces an algebraic multiplier s and subtracts the initial boundary condition value $f(0)$, while the integration property scales the transform by a factor of $\frac{1}{s}$ in the frequency domain.

Q8.99.1 | MCQ | 1999 | 1M | Ans: D | ● ● ○

We are given a constant pulse function $f(t)$ defined as:

$$f(t) = \begin{cases} k & \text{if } 0 < t < c \\ 0 & \text{if } c < t < \infty \end{cases}$$

We need to find its Laplace transform.
Let us express this pulse algebraically using unit step functions ($u(t)$):

$$f(t) = k\Big[u(t) - u(t-c)\Big]$$

Now, take the Laplace transform of both terms using the time-shifting property ($\mathcal{L}\{u(t-c)\} = \frac{e^{-cs}}{s}$):

$$\mathcal{L}\{f(t)\} = k\left(\frac{1}{s}\right) - k\left(\frac{e^{-cs}}{s}\right) = \frac{k}{s}(1 - e^{-cs})$$

This corresponds to choice D.

$$\boxed{\text{D. } (k/s)(1 - e^{-cs})}$$

Key Points

- A constant pulse spanning from 0 to c with a height of k is modeled as a combination of two step functions. Its Laplace transform evaluates directly to $\frac{k(1-e^{-cs})}{s}$.

Q8.98.1 | MCQ | 1998 | 1M | Ans: A | ● ○ ○

We are given a delayed unit step function $u_a(t)$ defined as:

$$u_a(t) = \begin{cases} 0 & \text{for } t < a \\ 1 & \text{for } t > a \end{cases}$$

We need to state its standard Laplace transform representation.
By standard definition of the unilateral Laplace integral:

$$\mathcal{L}\{u_a(t)\} = \int_a^\infty e^{-st}(1)\, dt = \left[\frac{e^{-st}}{-s}\right]_a^\infty = 0 - \left(\frac{e^{-as}}{-s}\right) = \frac{e^{-as}}{s}$$

This corresponds to choice A.

$$\boxed{\text{A. } e^{-as}/s}$$

Key Points

- Delaying a unit step function by a time interval a shifts the lower integration boundary, which multiplies the standard transform $(\frac{1}{s})$ by an exponential factor e^{-as} in the frequency domain.

Q8.96.1 MCQ 1996 1M Ans: D ● ● ○

We need to find the inverse Laplace transform of the function:

$$F(s) = (s+1)^{-2} = \frac{1}{(s+1)^2}$$

Let us use the repeated root property of the inverse Laplace transform along with the frequency shifting theorem. We know the standard transform pair for a repeated root at the origin is:

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} = t$$

According to the First Shifting Theorem, replacing s with $(s+1)$ shifts the expression in the frequency domain, which is equivalent to multiplying the time-domain function by an exponential factor $e^{-1\cdot t}$:

$$\mathcal{L}^{-1}\left\{\frac{1}{(s+1)^2}\right\} = t \cdot e^{-t}$$

This matches choice D perfectly.

$$\boxed{\text{D. } te^{-t}}$$

Key Points

- Rational functions containing a repeated pole root factor $(s+a)^{-n}$ are transformed back to the time domain using the shifting theorem combined with the standard power pair: $\mathcal{L}^{-1}\left\{\frac{1}{(s+a)^n}\right\} = \frac{t^{n-1}}{(n-1)!}e^{-at}$.

Q8.95.1 MCQ 1995 1M Ans: D ● ● ○

We need to find the inverse Laplace transform of the rational function:

$$F(s) = \frac{s+9}{s^2+6s+13}$$

First, let us rewrite the quadratic polynomial in the denominator by completing the square:

$$s^2 + 6s + 13 = (s^2 + 6s + 9) + 4 = (s+3)^2 + 2^2$$

Now, rearrange the terms in the numerator to match the shifted factor $(s+3)$ from the denominator:

$$s + 9 = (s+3) + 6$$

Substitute these expressions back into $F(s)$ and split it into two separate fractional terms:

$$F(s) = \frac{(s+3)+6}{(s+3)^2+2^2} = \frac{s+3}{(s+3)^2+2^2} + \frac{6}{(s+3)^2+2^2}$$

$$F(s) = \frac{s+3}{(s+3)^2+2^2} + 3\cdot\left[\frac{2}{(s+3)^2+2^2}\right]$$

Apply the inverse Laplace transform term-by-term using the First Shifting Theorem combined with standard cosine and sine pairs ($\mathcal{L}\{\cos \omega t\} = \frac{s}{s^2 + \omega^2}$ and $\mathcal{L}\{\sin \omega t\} = \frac{\omega}{s^2 + \omega^2}$): The shared shift factor $(s + 3)$ multiplies both time-domain terms by an exponential factor e^{-3t}:

$$f(t) = e^{-3t} \cos(2t) + 3e^{-3t} \sin(2t)$$

This corresponds to choice D.

$$\boxed{\text{D. } e^{-3t} \cos 2t + 3e^{-3t} \sin 2t}$$

SOLUTIONS — Transform Theory

CH

Chemical Engineering

Q8.20.1 MCQ | 2020 | 1M | Ans: A | ● ● ○

We are given a delayed unit step function $f(t)$ as shown ;

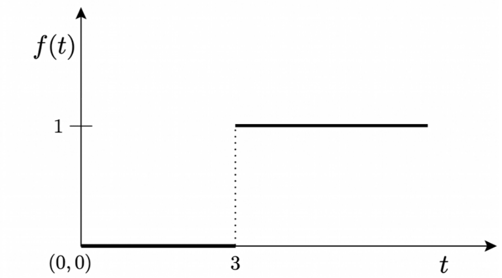

Fig. Solution Q8.20.1

$$f(t) = \begin{cases} 0 & \text{if } 0 \le t < 3 \\ 1 & \text{if } t \ge 3 \end{cases}$$

This can be written in standard notation as $f(t) = u(t-3)$, where the variable s is represented as x in the choices. We need to find its Laplace transform.
Applying the standard canonical integral definition of the unilateral Laplace transform:

$$\mathcal{L}\{u(t-3)\} = \int_3^\infty e^{-xt} \cdot (1)\, dt = \left[\frac{e^{-xt}}{-x}\right]_3^\infty = 0 - \left(\frac{e^{-3x}}{-x}\right) = \frac{e^{-3x}}{x}$$

This corresponds to choice A.

$$\boxed{\text{A. } \frac{e^{-3x}}{x}}$$

Key Points

- A unit step function delayed by a constant time interval c introduces an exponential decay scaling factor e^{-cs} in the Laplace domain: $\mathcal{L}\{u(t-c)\} = \dfrac{e^{-cs}}{s}$.

Q8.17.1 MCQ | 2017 | 2M | Ans: D | ● ● ○

We are given the Laplace transform of a continuous time signal, with the complex frequency variable written as x:

$$F(x) = \frac{x+1}{x(x+2)}$$

We need to determine the initial value $f(0)$ and final value $f(\infty)$ of the function, respectively.
Step 1: Calculate the initial value $f(0)$ using the Initial Value Theorem (IVT)

$$f(0) = \lim_{x\to\infty} xF(x) = \lim_{x\to\infty} x\left[\frac{x+1}{x(x+2)}\right] = \lim_{x\to\infty} \frac{x+1}{x+2}$$

Divide the numerator and denominator by x:

$$f(0) = \lim_{x\to\infty} \frac{1+\frac{1}{x}}{1+\frac{2}{x}} = \frac{1+0}{1+0} = 1$$

Step 2: Calculate the final value $f(\infty)$ using the Final Value Theorem (FVT) First, let us verify stability by locating the system poles (roots of the denominator polynomial):

$$x(x+2) = 0 \implies x = 0 \text{ and } x = -2$$

The pole at the origin $x = 0$ is simple, and the other pole lies strictly in the left half of the complex plane ($\text{Re}(x) = -2 < 0$). This ensures the system is stable over time, validating the use of the FVT:

$$f(\infty) = \lim_{x\to 0} xF(x) = \lim_{x\to 0} x\left[\frac{x+1}{x(x+2)}\right] = \lim_{x\to 0}\frac{x+1}{x+2} = \frac{0+1}{0+2} = \frac{1}{2}$$

Thus, the initial and final values are 1 and $\frac{1}{2}$, respectively. This corresponds to choice B.

$$\boxed{\text{B: } 1 \; and \; \frac{1}{2}}$$

Key Points

- The Initial Value Theorem is computed using $\lim_{s\to\infty} sF(s)$, while the Final Value Theorem evaluates to $\lim_{s\to 0} sF(s)$, provided all poles of $sF(s)$ lie strictly in the left half of the complex s-plane.

Q8.16.1 MCQ 2016 1M Ans: A ● ● ○

We need to identify the Laplace transform of the exponentially modulated sine function, where the s-domain parameter is labeled as x:

$$f(t) = e^{at}\sin(bt)$$

According to the First Shifting Theorem, multiplying any time-domain signal by e^{at} translates its frequency representation by shifting the variable $x \to (x-a)$.
First, state the canonical transform of the core sine function $g(t) = \sin(bt)$:

$$G(x) = \mathcal{L}\{\sin(bt)\} = \frac{b}{x^2+b^2}$$

Now, substitute x with $(x-a)$ to account for the exponential multiplier e^{at}:

$$\mathcal{L}\{e^{at}\sin(bt)\} = G(x-a) = \frac{b}{(x-a)^2+b^2}$$

This matches choice A perfectly.

$$\boxed{\text{A. } \frac{b}{(x-a)^2+b^2}}$$

Key Points

- **First Shifting Theorem**: Frequency domain translation replaces the complex variable x with a shifted parameter $(x-a)$ whenever the time function is modulated by an exponential factor e^{at}.

Q8.14.1 MCQ 2014 1M Ans: D ● ● ○

We are given the time-domain function $f(t) = t^2$, with the frequency variable denoted as x. We need to find the Laplace transform of its definite integral:

$$g(t) = \int_0^t f(\tau)\,d\tau = \int_0^t \tau^2\,d\tau$$

Let us utilize the standard integration property of the Laplace transform:

$$\mathcal{L}\left\{\int_0^t f(\tau)\,d\tau\right\} = \frac{1}{x}F(x)$$

First, find the Laplace transform of the core polynomial term $f(t) = t^2$ using the standard pair $\mathcal{L}\{t^n\} = \dfrac{n!}{x^{n+1}}$:

$$F(x) = \mathcal{L}\{t^2\} = \frac{2!}{x^{2+1}} = \frac{2}{x^3}$$

Now, apply the integration property by dividing this result by x:

$$\mathcal{L}\left\{\int_0^t \tau^2\,d\tau\right\} = \frac{1}{x}\cdot F(x) = \frac{1}{x}\cdot\left(\frac{2}{x^3}\right) = \frac{2}{x^4}$$

This corresponds to choice D.

$$\boxed{\text{D. } \frac{2}{x^4}}$$

Key Points

- **Integration Property**: Definite integration of a causal signal from 0 to t is equivalent to multiplying its Laplace transform by a factor of $\frac{1}{s}$ (or $\frac{1}{x}$) in the frequency domain.

Q8.12.1 MCQ 2012 1M Ans: D ● ● ○

We are given that the unilateral Laplace transform of a function $f(t)$ is:

$$F(x) = \frac{1}{x^2 + x + 1}$$

We need to find the Laplace transform of the time-multiplied function $g(t) = tf(t)$.
Let us utilize the frequency domain differentiation property of the Laplace transform:

$$\mathcal{L}\{t\cdot f(t)\} = -\frac{d}{dx}\Big[F(x)\Big]$$

Let us compute the derivative of $F(x)$ with respect to x using the chain rule:

$$\frac{d}{dx}\left[\left(x^2+x+1\right)^{-1}\right] = -1\cdot\left(x^2+x+1\right)^{-2}\cdot\frac{d}{dx}\left(x^2+x+1\right)$$

$$\frac{d}{dx}\Big[F(x)\Big] = \frac{-(2x+1)}{\left(x^2+x+1\right)^2}$$

Now, multiply by the negative sign required by the property:

$$\mathcal{L}\{t\cdot f(t)\} = -\left[\frac{-(2x+1)}{\left(x^2+x+1\right)^2}\right] = \frac{2x+1}{\left(x^2+x+1\right)^2}$$

$$\boxed{\text{D. } \frac{(2x+1)}{(x^2+x+1)^2}}$$

Key Points

- **Frequency Differentiation Theorem**: Multiplying a time-domain signal by t corresponds to differentiating its Laplace transform with respect to the frequency variable and inverting the sign: $\mathcal{L}\{tf(t)\} = -\frac{d}{ds}F(s)$.

Q8.08.1 | MCQ | 2008 | 2M | Ans: A | ● ● ○

We need to find the Laplace transform of the modulated time function:

$$f(t) = t\sin(t)$$

where the frequency domain parameter is written as x.
Let us apply the frequency domain differentiation property:

$$\mathcal{L}\{t \cdot \sin(t)\} = -\frac{d}{dx}\Big[G(x)\Big]$$

where $G(x)$ is the Laplace transform of the core sine term $g(t) = \sin(t)$ with $\omega = 1$:

$$G(x) = \mathcal{L}\{\sin(t)\} = \frac{1}{x^2 + 1^2} = \frac{1}{x^2 + 1}$$

Now, compute the derivative of $G(x)$ with respect to x using the quotient rule or chain rule:

$$\frac{d}{dx}\left[\left(x^2 + 1\right)^{-1}\right] = -1 \cdot \left(x^2 + 1\right)^{-2} \cdot (2x) = \frac{-2x}{\left(x^2 + 1\right)^2}$$

Multiply by the negative sign required by the property:

$$\mathcal{L}\{t \cdot \sin(t)\} = -\left[\frac{-2x}{\left(x^2 + 1\right)^2}\right] = \frac{2x}{\left(x^2 + 1\right)^2}$$

This matches choice A perfectly.

$$\boxed{\text{A. } \frac{2x}{(x^2 + 1)^2}}$$

Key Points

- Multiplying a function by the time variable t is equivalent to differentiating its Laplace transform in the frequency domain: $\mathcal{L}\{t\sin(\omega t)\} = \frac{2\omega s}{(s^2 + \omega^2)^2}$.

Q8.07.1 | MCQ | 2007 | 2M | Ans: A | ● ● ○

We need to find the Laplace transform of the fractional radical power function:

$$f(t) = \frac{1}{\sqrt{t}} = t^{-1/2}$$

where the transform variable is written as x.
Let us apply the generalized polynomial power formula for the Laplace transform, which utilizes the Gamma function (Γ):

$$\mathcal{L}\{t^n\} = \frac{\Gamma(n + 1)}{x^{n+1}}$$

Substitute $n = -\frac{1}{2}$ into this formula:

$$\mathcal{L}\left\{t^{-1/2}\right\} = \frac{\Gamma\left(-\frac{1}{2}+1\right)}{x^{-\frac{1}{2}+1}} = \frac{\Gamma\left(\frac{1}{2}\right)}{x^{1/2}}$$

Using the standard mathematical value for the Gamma function at a half-integer, $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$:

$$\mathcal{L}\left\{t^{-1/2}\right\} = \frac{\sqrt{\pi}}{\sqrt{x}} = \sqrt{\frac{\pi}{x}}$$

This corresponds to choice A.

$$\boxed{\text{A. } \sqrt{\frac{\pi}{x}}}$$

Key Points

- **Fractional Power Identity**: For non-integer power functions, use the Gamma function expansion. The transform of $t^{-1/2}$ evaluates precisely to the closed radical expression $\sqrt{\frac{\pi}{s}}$.

Q8.98.1 MCQ 1998 1M Ans: D ● ○ ○

We need to state the standard algebraic form of the unilateral Laplace transform of the exponential decay function:

$$f(t) = e^{-at}$$

where the transform variable is denoted as x.
By standard definition of the unilateral Laplace integral:

$$\mathcal{L}\{e^{-at}\} = \int_0^\infty e^{-xt} \cdot e^{-at}\, dt = \int_0^\infty e^{-(x+a)t}\, dt$$

Evaluate the definite integral for $\mathrm{Re}(x) > -a$:

$$\mathcal{L}\{e^{-at}\} = \left[\frac{e^{-(x+a)t}}{-(x+a)}\right]_0^\infty = 0 - \left(\frac{1}{-(x+a)}\right) = \frac{1}{x+a}$$

This corresponds to choice D.

$$\boxed{\text{D. } \frac{1}{x+a}}$$

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Q8.12.1 MCQ 2012 1M Ans: D ● ● ○

We need to find the inverse Laplace transform of the function:

$$F(s) = \frac{1}{s(s+1)}$$

Let us expand $F(s)$ into partial fractions using the standard linear decomposition template:

$$\frac{1}{s(s+1)} = \frac{A}{s} + \frac{B}{s+1}$$

Using the residue cover-up method to evaluate the constants:

- For A (at $s = 0$): $A = \left.\frac{1}{s+1}\right|_{s=0} = \frac{1}{0+1} = 1$
- For B (at $s = -1$): $B = \left.\frac{1}{s}\right|_{s=-1} = \frac{1}{-1} = -1$

Substitute these coefficients back into the partial fraction expansion lines:

$$F(s) = \frac{1}{s} - \frac{1}{s+1}$$

Now, apply the linear inverse Laplace transform term-by-term using standard transforming pairs ($\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = 1$ and $\mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\} = e^{-t}$):

$$f(t) = 1 - e^{-t}$$

This corresponds to choice D.

$$\boxed{\text{D. } f(t) = 1 - e^{-t}}$$

Key Points

- Rational functions with simple real roots can be decomposed into standard partial fractions to directly compute inverse transforms term-by-term.

Q8.08.1 MCQ 2008 1M Ans: A ● ○ ○

We need to find the standard canonical Laplace transform of the hyperbolic sine function:

$$f(t) = \sinh(ht)$$

where the scaling factor is given as $h = 1$, modifying the function to $\sinh(t)$. Let us state the standard exponential definition of the hyperbolic sine function:

$$\sinh(at) = \frac{e^{at} - e^{-at}}{2}$$

Applying the unilateral integral or using standard operational transform parameters with $a = 1$:

$$\mathcal{L}\{\sinh(at)\} = \frac{a}{s^2 - a^2}$$

Substituting $a = 1$ directly into the algebraic pair formula yields:

$$\mathcal{L}\{\sinh(t)\} = \frac{1}{s^2 - 1^2} = \frac{1}{s^2 - 1}$$

This matches choice A perfectly.

$$\boxed{\text{A. } \frac{1}{s^2 - 1}}$$

Key Points

- The unilateral Laplace transform maps the fundamental hyperbolic function $\sinh(at)$ directly to the expression $\frac{a}{s^2 - a^2}$ for $\text{Re}(s) > |a|$.

Q8.08.2 MCQ 2008 1M Ans: D ● ● ○

We need to find the Laplace transform of the polynomial monomial function:

$$f(t) = 8t^3$$

Let us utilize the linearity property of the Laplace transform to factor out the constant coefficient:

$$\mathcal{L}\{8t^3\} = 8 \cdot \mathcal{L}\{t^3\}$$

Now, apply the standard integer power function transform pair formula:

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$$

Substituting the exponent parameter value $n = 3$:

$$\mathcal{L}\{t^3\} = \frac{3!}{s^{3+1}} = \frac{3 \times 2 \times 1}{s^4} = \frac{6}{s^4}$$

Multiply this result by the external scalar constant factor 8:

$$\mathcal{L}\{8t^3\} = 8 \times \left(\frac{6}{s^4}\right) = \frac{48}{s^4}$$

This calculation corresponds to choice D.

$$\boxed{\text{D. } \frac{48}{s^4}}$$

Key Points

- Polynomial power transformations scale as a factorial function of their degree in the numerator divided by the frequency parameter raised to power $n + 1$ in the denominator.

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Engineering Mathematics

Total Questions : 43

MCQ : 42

MSQ : 0

NAT : 1